FORENSIC ANALYSIS REPORT

Blood Pattern Analysis & Classification

Report Generated: March 10, 2026 at 12:13 PM

EXECUTIVE SUMMARY

Analysis Component	Result	Confidence
Blood Detection	■ LIKELY BLOOD	80.7%
Weapon Type	Gun	100.0%
Point of Origin	(256.9, 248.7)	400 droplets

1. BLOOD DETECTION ANALYSIS

Verdict: ■ LIKELY BLOOD

Confidence: 80.7%

Color Analysis:

Coverage: 21.12%

Score: 70.4%

Detected Blood Types:

• Fresh Blood (Bright Red): 14.64%

• Fresh Blood (Deep Red): 5.72%

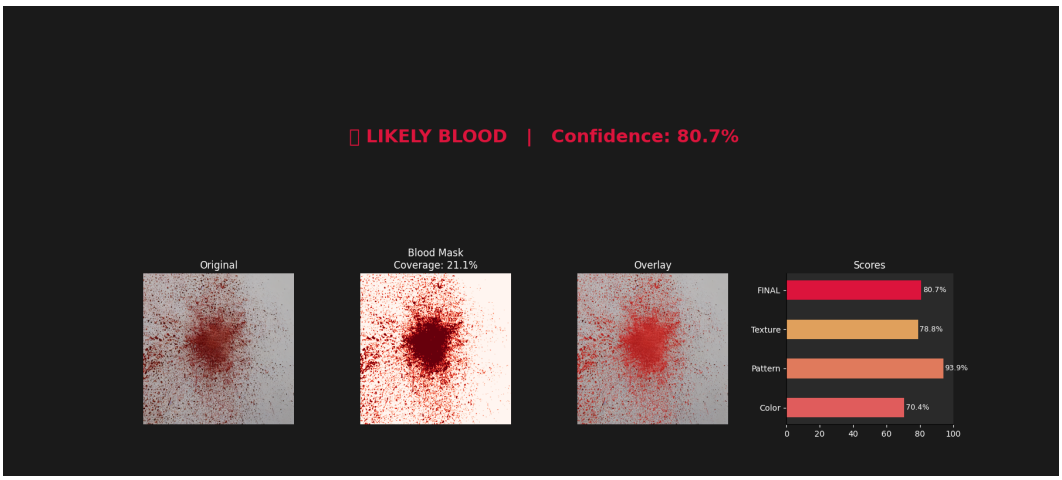
• Dried Blood (Brown-Red): 4.26%

Pattern Analysis:

Score: 93.9%

Components: 443

Satellites: 442



Blood Detection Visualization

2. WEAPON TYPE CLASSIFICATION

Weapon Type: Gun

Confidence: 100.0%

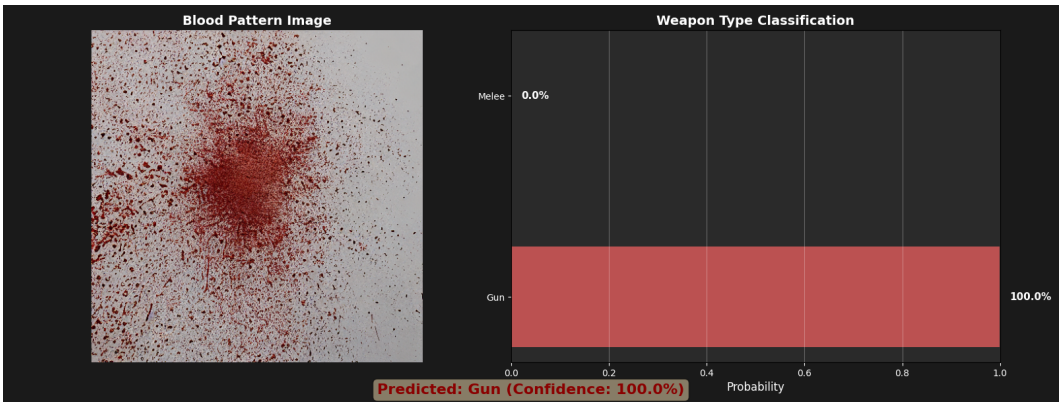
Detailed Probabilities:

• Gun: 99.99%

• Melee: 0.01%

Interpretation:

High/medium velocity impact spatter detected. Characteristics: Fine mist, small circular droplets, uniform distribution typical of gunshot wounds.



Weapon Classification Visualization

3. POINT OF ORIGIN ANALYSIS

Point of Origin: (256.9, 248.7)

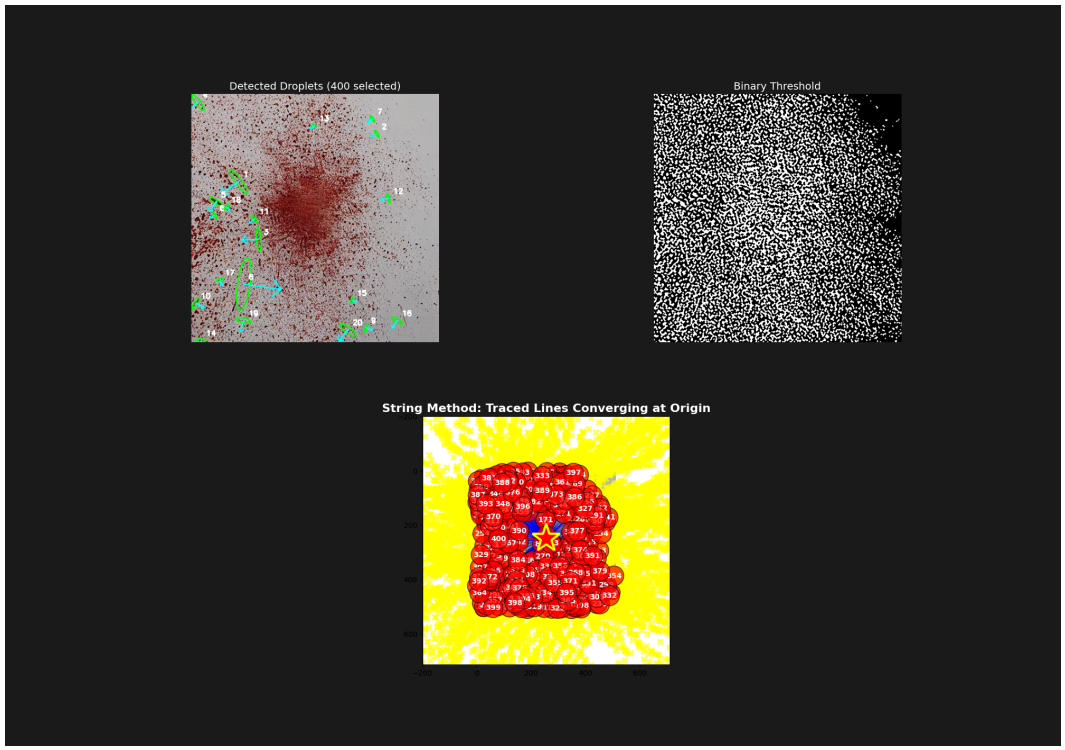
Analysis Statistics:

Droplets Analyzed: 400

Intersection Points: 38623

Average Impact Angle: 34.5°

Angle Range: 12.5° - 56.4°



String Method Visualization

DISCLAIMER

This forensic analysis report was generated using AI-assisted computational methods for blood pattern analysis. The results presented are based on image analysis algorithms and machine learning models trained on forensic datasets. **Important Notes:**

• This report should be used as a supplementary tool in forensic investigations
• Results must be verified by qualified forensic experts
• This report is not intended for sole use in legal proceedings
• Professional forensic confirmation is required for official documentation
• The accuracy of results depends on image quality and pattern clarity **Model Information:**
• Blood Detection: Computer vision analysis (HSV color space, pattern recognition)
• Weapon Classification: Deep learning CNN (TensorFlow)
• String Method: Classical forensic trajectory analysis with DBSCAN clustering For questions or concerns about this analysis, consult with certified forensic analysts.